

ANALYSTLAB AFRICA — WEEK 1–2 DATA CLEANING & EDA REPORT

Online Retail E-commerce Dataset

PROJECT OBJECTIVE

To clean, validate, and analyze raw transactional data from the Online Retail dataset by handling data quality issues and extracting key business insights through exploratory data analysis (EDA).

DATA CLEANING SUMMARY

- Removed records with missing CustomerID for valid customer analysis
- Eliminated duplicate transactions to avoid inflated metrics
- Filtered invalid values in Quantity and UnitPrice (≤ 0)
- Standardized column names for consistency
- Converted InvoiceDate to datetime format
- Created Revenue feature (Quantity $\times$ UnitPrice)

EXPLORATORY DATA ANALYSIS (EDA)

- Identified top-selling products by quantity
- Analyzed revenue contribution by country
- Evaluated monthly sales trends and seasonality
- Determined top customers by revenue contribution
- Generated summary statistics for key numerical fields

VISUALIZATIONS CREATED

- Top countries by revenue (bar chart)
- Monthly revenue trend (line chart)
- Top-selling products (bar chart)
- Customer revenue distribution (histogram)
- Unit price distribution and outliers (box plot)

KEY INSIGHTS

- United Kingdom dominates total revenue
- Revenue follows a strong Pareto pattern (few customers generate most sales)
- Sales show seasonal fluctuations across months
- A small number of products drive most transactions
- Customer behavior is split between frequent low-value and rare high-value purchases

CONCLUSION

The dataset was successfully cleaned and structured for analysis. Key insights into customer behavior, product performance, and revenue distribution were identified, forming a foundation for forecasting and customer segmentation.